

Dr. Sabrina Amri

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PROFILE

Ph.D. in Computer Science specializing in cybersecurity applied to artificial intelligence, with expertise in disinformation detection, privacy of AI models, explainable artificial intelligence (XAI), large language models (LLMs), machine learning, and multimodal analysis. Experienced in advanced research on security, privacy, and disinformation risks in artificial intelligence systems and social networks.

EDUCATION

Ph.D. in Computer Science 2019 – 2024

Université de Montréal, Canada

- Thesis: “FACTS-ON: Fighting Against Counterfeit Truths in Online Social Networks: Fake News, Misinformation, and Disinformation.”
- Field: Detection and mitigation of false information using AI techniques, including transformers, explainable AI, and multimodal analysis.

Master's in Intelligent and Pervasive Systems 2015 – 2018

Institut Supérieur des Sciences Appliquées et de Technologie de Sousse, Tunisia

National Engineering Diploma in Computer Science 2012 – 2015

École Polytechnique de Sousse, Tunisia

Bachelor's Degree in Network and Services Administration 2007 – 2010

Institut Supérieur des Sciences Appliquées et de Technologie de Sousse, Tunisia

ADDITIONAL TRAINING

Machine Learning for Life Sciences March 2025

Université de Montréal – IVADO | In-person continuing education (8h)

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher in Cybersecurity and Artificial Intelligence 2025 – Present

Polytechnique Montréal, Canada

Research on privacy, security, and explainability of large language models (LLMs).

- Development of neuro-symbolic and multi-agent systems for mitigating privacy risks in generative AI.
- Work on disinformation detection, digital content provenance, and trustworthy AI.
- Participation in cybersecurity research projects related to emerging risks of artificial intelligence.

Cybersecurity and Artificial Intelligence Researcher 2019 – 2024

Université de Montréal, Canada

Design and development of several advanced fake news detection systems integrating textual content, images, social context, and external evidence, with emphasis on explainability (XAI) and multimodal analysis (text and images).

- Development of EXMULF, an explainable platform based on ViBERT and LIME for multimodal fake news detection from social media posts.
- Creation of ExFake, a system combining NLP tasks (NER, NLI), web scraping techniques, and contextual analyses to identify disinformation with 90% accuracy.
- Development of AFCC, a framework for automatic inference of consensus among fact-checkers, using K-means clustering and Bayesian models to assess the credibility of verifications.
- Design of a disinformation classifier based on users' online personality, leveraging the Big Five model and a DistilBERT neural network, achieving 96% accuracy and outperforming five state-of-the-art models.
- Collection and processing of large-scale data from platforms such as Twitter and Weibo, in collaboration with interdisciplinary research teams (engineers, data scientists).

Teaching Assistant

2019 – 2023

Université de Montréal, Canada

- IFT1810: Introduction to Programming (C and Java). IFT1135: Introduction to Mobile Applications.
- Led lab sessions and taught programming fundamentals in C, Java, and Android; presented tutorials, supervised exams, graded assignments, and mentored students.

Software Developer Intern

2017

Department of Telecommunications Engineering, School of Engineering, Linares, Jaén, Spain

Software Developer Intern

2015

Laboratory of Electronics and Microelectronics, Faculty of Sciences, Monastir, Tunisia

- Design of optimization algorithms for virtual machine placement in cloud computing environments.
- Design and development of web and mobile applications for medical assistance.

Coordinator

2011 – 2012

Center for Strategic Studies and Development, Sidi Bouzid, Tunisia

- Team management, event coordination, and direction of community projects in collaboration with several organizations.

Coordination and Information Systems Intern

2011

Department of Coordination of Saône-et-Loire, France

IT Support Intern

2009 – 2010

Tunisie Telecom, Sousse, Tunisia

- Development of an FTP application for telecommunications.

SCIENTIFIC COMMITTEES AND EVALUATION

NSERC Scholarship Evaluation Committee Member

2025 – 2026

- Selection committee member – NSERC Scholarships (Doctoral) 2025.
- Selection committee member – NSERC Scholarships (Master's) 2026.

- Academic and scientific evaluation of multidisciplinary applications in engineering, including computer science, electronics, mechanical engineering, medicine, and bioinformatics.
- Review of applications according to criteria of academic excellence, research potential, scientific innovation, and interdisciplinary impact.

Scientific Peer Review Activities

- Scientific reviewer for several international journals and conferences in computer science, artificial intelligence, cybersecurity, and social network analysis.
- Evaluation of scientific manuscripts on machine learning, online disinformation, intelligent systems, and multimodal analysis.
- Journals including: ACM Computing Surveys – SN Computer Science – Online Social Networks and Media – International Journal of Disaster Risk Reduction

PUBLICATIONS

- [1] Sabrine Amri, Nora Boulahia-Cuppens, Frédéric Cuppens. "PRISM-AI: A Dual-Stage Neuro-Symbolic Agentic Framework for Privacy Risk Mitigation in LLMs." 2025 IEEE/ACS 22nd International Conference on Computer Systems and Applications (AICCSA), (pp. 01-07). IEEE, 2025.
- [2] Dorsaf Sallami, Sabrine Amri, Esma Aïmeur. "ClaimVerAgents: A Multi-Agent Retrieval-Augmented Claim Verification Framework." 2025 IEEE/ACS 22nd International Conference on Computer Systems and Applications (AICCSA), (pp. 01-06). IEEE, 2025.
- [3] Sabrine Amri, Esma Aïmeur. "AFCC: Towards an Automatic Fact-Checkers' Consensus Inference and Credibility Assessment for Trust-Based Fake News Detection." International Journal of Information Technology (2024): 1-16.
- [4] Sabrine Amri, Henri-Cedric Mputu Boleilanga, Esma Aïmeur. "ExFake: Towards an Explainable Fake News Detection Based on Content and Social Context Information." 2023 Congress in Computer Science, Computer Engineering & Applied Computing (CSCE), (pp. 01-08). IEEE.
- [5] Esma Aïmeur, Sabrine Amri, Gilles Brassard. "Fake News, Disinformation and Misinformation in Social Media: A Review." Social Network Analysis and Mining 13.1 (2023): 30.
- [6] Sabrine Amri, Dorsaf Sallami, Esma Aïmeur. "EXMULF: An Explainable Multimodal Content-Based Fake News Detection System." International Symposium on Foundations and Practice of Security. Springer, Cham, 2022.
- [7] Hedi Hamdi, Sabrine Amri, Zaki Brahmi. "Managing Performance Interference Effects for Intelligent and Efficient Virtual Machines Placement Based on GWO Approach in Cloud." International Journal of Computing and Digital Systems 8.04 (2019): 317-332.
- [8] Esma Aïmeur, Hicham Hage, Sabrine Amri. "The Scourge of Online Deception in Social Networks." 2018 International Conference on Computational Science and Computational Intelligence (CSCI). IEEE, 2018.
- [9] Sabrine Amri, Zaki Brahmi, Rocio P. Prado, S. Garcia-Galan, J. E. Muñoz-Expósito. "Interference-Aware Virtual Machine Placement: A Survey." International Conference on Image Processing and Communications Challenges. Springer, Cham, 2018.

[10] Sabrine Amri, Hedi Hamdi, Zaki Brahmi. "Inter-VM Interference in Cloud Environments: A Survey." 2017 IEEE/ACS 14th International Conference on Computer Systems and Applications (AICCSA). IEEE, 2017.

TECHNICAL SKILLS

- **Natural Language Processing (NLP):** BERT, Word2Vec, spaCy, LDA
- **Machine Learning and Deep Learning:** Scikit-Learn, Deep Learning (including PyTorch, ViBERT), LIME (model interpretability), K-means clustering
- **Software development and programming:** Pascal, C/C++, Java, Python, PHP5, HTML5, CSS3, JavaScript, Prolog, Lisp, MySQL, SQL, mobile development (Android), embedded development (Arduino, VHDL), MongoDB, JSON, Scrapy
- **Cisco Certifications:** Cisco CCNA1, 2, 3

LANGUAGES

Arabic: Native | French: Fluent | English: Fluent